SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1711**

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 20 **Duration:30 Minutes**
Total Marks:20

Annexure A
MCQ Test

Code of Conduct:

I SHABARISH N 192424007 (*Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

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1. In informed search strategies, some algorithms use only the heuristic value to decide which node to expand next. This approach focuses on selecting the node that appears closest to the goal without considering the cost already incurred. Which algorithm follows this principle?

A) Breadth First Search

B) Uniform Cost Search

C) Greedy Best-First Search ***✓Correct Answer***

D) Depth First Search

Explanation: *Greedy Best-First Search expands the node with the lowest heuristic value h(n) only, ignoring the cost already incurred (g(n)).*

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2. A* search is widely used in pathfinding and graph traversal because it combines the actual cost from the start node and the estimated cost to the goal. What is the evaluation function used in A* search?

A) $f(n) = g(n)$

B) $f(n) = h(n)$

C) $f(n) = g(n) + h(n)$ ***✓Correct Answer***

D) $f(n) = g(n) \times h(n)$

Explanation: *A* evaluates nodes using $f(n) = g(n) + h(n)$, the sum of the path cost so far and the estimated cost to the goal.*

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3. Heuristic functions play a crucial role in informed search by guiding the algorithm toward the goal. What property ensures optimality in A* search?

A) Completeness

B) Admissibility ***✓Correct Answer***

C) Depth limit

D) Backtracking

Explanation: *An admissible heuristic never overestimates the true cost to the goal, which guarantees A* finds the optimal solution.*

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4. Local search algorithms operate by starting with an initial solution and iteratively improving it based on a given objective function. Which of the following is a local search algorithm?

A) A* Search

B) Hill Climbing ***✓Correct Answer***

C) Breadth First Search

D) Uniform Cost Search

Explanation: *Hill Climbing iteratively moves to a better neighboring state based on an objective function, without tracking a path — a classic local search method.*

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5. In adversarial search, agents compete against each other in a game environment, considering both the agent's moves and the opponent's possible responses. Which algorithm is commonly used for such decision-making?

A) Greedy Search

B) Minimax Algorithm ***✓Correct Answer***

C) Backtracking

D) Uniform Search

Explanation: *Minimax evaluates game states by assuming the opponent plays optimally, alternating between maximizing and minimizing utility.*

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6. A delivery robot in a warehouse selects paths based only on the estimated distance to the destination without considering obstacles already encountered. Which algorithm is the robot using?

A) A* Search

B) Greedy Best-First Search ✓*Correct Answer*

C) Uniform Cost Search

D) Depth Limited Search

Explanation: *Using only the heuristic estimate to the goal (ignoring cost so far) is the defining behavior of Greedy Best-First Search.*

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7. A GPS navigation system calculates routes by combining the distance already travelled and an estimate of the remaining distance to the destination. Which algorithm best represents this approach?

A) Depth First Search

B) A* Search ✓*Correct Answer*

C) Hill Climbing

D) Backtracking

Explanation: *Combining cost-so-far with an estimate of remaining cost is exactly $f(n) = g(n) + h(n)$, the A* search formulation.*

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8. A university is scheduling exams for multiple subjects with constraints such as no overlapping exams and limited rooms. This problem is best modeled as:

A) Game Playing Problem

B) Constraint Satisfaction Problem ✓*Correct Answer*

C) Heuristic Search Problem

D) Optimization Problem

Explanation: *Assigning values (time slots) to variables (exams) subject to constraints (no overlap, room limits) is a textbook Constraint Satisfaction Problem.*

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9. An AI-based chess program evaluates all possible moves and anticipates the opponent's responses before making a decision, assigning utility values to game states. Which algorithm is being applied here?

A) Greedy Search

B) Hill Climbing

C) Minimax Algorithm ✓*Correct Answer*

D) Breadth First Search

Explanation: *Assigning utility values to states while anticipating an opponent's optimal replies is the Minimax approach to adversarial games.*

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10. A robot explores an unknown environment where it does not have prior knowledge of obstacles. It makes decisions step-by-step and updates its knowledge as it moves. Which type of agent is suitable for this scenario?

A) Offline Agent

B) Online Search Agent ✓*Correct Answer*

C) Static Agent

D) Deterministic Agent

Explanation: *An Online Search Agent acts and senses the environment interleaved, updating its knowledge as it moves through an initially unknown space.*

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11. A* search guarantees optimal solutions only when certain conditions are satisfied by the heuristic function. Why is an admissible heuristic important?

A) It reduces memory usage

B) It ensures optimality ✓*Correct Answer*

C) It increases branching factor

D) It avoids loops

Explanation: *An admissible heuristic never overestimates the true cost, which is the condition required for A* to guarantee an optimal solution.*

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12. In large game trees, evaluating every node becomes computationally expensive. What is the primary benefit of Alpha-Beta pruning?

A) Reduces depth of tree

B) Reduces number of nodes evaluated ✓*Correct Answer*

C) Changes game rules

D) Improves heuristic accuracy

Explanation: *Alpha-Beta pruning cuts off branches that cannot influence the final decision, reducing the number of nodes minimax needs to evaluate.*

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13. Hill Climbing algorithm continuously moves toward better states based on local improvements. What is the main drawback of Hill Climbing?

A) High memory usage

B) Revisiting nodes

C) Getting stuck in local maxima ✓*Correct Answer*

D) Requires complete tree

Explanation: *Because Hill Climbing only accepts moves that locally improve the state, it can get trapped at a local maximum that isn't the global best.*

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14. Constraint Satisfaction Problems involve variables, domains, and constraints. What ensures a valid CSP solution?

A) Heuristic function

B) Path cost

C) Constraint satisfaction ✓*Correct Answer*

D) Depth-first traversal

Explanation: *A valid CSP solution is an assignment of values to all variables that satisfies every constraint — i.e., constraint satisfaction itself.*

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15. In adversarial games, evaluation functions are used when it is not feasible to explore the entire game tree. What is the role of an evaluation function?

A) Generate successors

B) Estimate utility of a state ✓*Correct Answer*

C) Reduce branching factor

D) Store moves

Explanation: *An evaluation function estimates how desirable (good) a non-terminal game state is, standing in for a full minimax expansion.*

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16. In A* search, the evaluation function is calculated using both the path cost and heuristic value. Suppose a node has a path cost of 7 and a heuristic estimate of 5. What is the value of f(n)?

- **A) 12** ✓***Correct Answer***
- B) 2
- C) 35
- D) 7

Explanation: *$f(n) = g(n) + h(n) = 7 + 5 = 12$.*

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17. A heuristic function sometimes overestimates the actual cost to reach the goal. This violates the admissibility condition required for optimal solutions in A* search. What happens in such a case?

- A) A* remains optimal
- **B) A* becomes non-optimal** ✓***Correct Answer***
- C) A* becomes BFS
- D) A* stops working

Explanation: *If h(n) overestimates the true cost, the admissibility condition is violated and A* is no longer guaranteed to find the optimal solution.*

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18. Consider a search tree with a branching factor of 2 and depth of 4. What is the approximate number of nodes in the full tree?

- A) 10
- B) 15
- **C) 31** ✓***Correct Answer***
- D) 16

Explanation: *Total nodes = $2^0+2^1+2^2+2^3+2^4 = 1+2+4+8+16 = 31$.*

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19. In a minimax tree, each node has 3 children and the depth of the tree is 3. How many leaf nodes are present?

- A) 9
- **B) 27** ✓***Correct Answer***
- C) 6
- D) 12

Explanation: *Leaf nodes = branching factor^depth = $3^3 = 27$.*

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20. Alpha-Beta pruning improves the efficiency of minimax by reducing the number of nodes evaluated. What is the best-case time complexity?

- A) O(b^d)
- **B) O(b^(d/2))** ✓***Correct Answer***
- C) O(d^b)
- D) O(log b)

Explanation: *In the best case (perfect move ordering), Alpha-Beta pruning reduces the effective branching factor, giving time complexity O(b^(d/2)).*